

IDENTIFICATION

CODE : GI-5-S1-EC-OTL
ECTS : 3

HOURS

Cours : 0h
TD : 24h
TP : 0h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 26h
Travail personnel : 30h
Total : 60h

ASSESSMENT METHOD

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Industrie 4.0 (GI-5-S1-UE-INQZ) and contributes to the following skills :

A2 Operate a model of a real or virtual system (Level 2)
C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
C5 Managing a production system and react to malfunctions (Level 2)
C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 1)
C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 1)
C13 Considering technological and methodological innovation (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

To be able to :

CONTENT

The engineering student will work and be assessed on the following knowledge and skills :

- (10h) Know the "classical" transportation problems : Traveling Salesman Problem (TSP), Vehicle Routing Problem (VRP), etc.
- (2h) Know the basic methods and tools to design logistic networks.
- (8h) Know the principles and tools for distributed decision-making: game theory, multi-agent systems and agent-based simulation coupled with flow simulation (for instance with Anylogic)
- (4h) Know the tools (especially decomposition methods) to perform joint optimisation.
- Be able to formulate and solve a supply chain optimisation problem, from the viewpoint of transportation and flow management.

BIBLIOGRAPHY

Ilya Grigoryev. AnyLogic 7 in Three Days : A Quick Course in Simulation Modeling. 3rd edition.
Jacques Teghem. Recherche opérationnelle. Tome 1, Tome 1, OCLC : 819196866. Paris : Ellipses, 2012. isbn : 978-2-7298-7509-1.
Paolo Toth et Daniele Vigo, eds. Vehicle routing : problems, methods, and applications. Second edition. MOS-SIAM series on optimization. Philadelphia : Society for Industrial et Applied Mathematics : Mathematical Optimization Society, 2014. 463 p. isbn : 978-1-61197-358-7.
Michael Wooldridge. An Introduction to Multi-Agent Systems. 2d edition. New York, NY, USA: John Wiley & Sons, Inc., 2009.
Murat Yildizoglu. Introduction à la Théorie des Jeux. Dunod, Paris (France), 2003.

PRE-REQUISITES

Inventory management : EOQ, basic replenishment methods
Flow management
Production management : S&OP, MPS & MRP
Scheduling: models, standard scheduling notation, heuristic solving methods for scheduling problems
Combinatorial optimisation: graph theory
Logistic